

**PREDICTIVE ANALYTICS FOR EMERGENCY DEPARTMENT (ED) MEDICINE AND
EQUIPMENT DEMAND FORECASTING USING PATIENT FLOW TRENDS AND
HISTORICAL USAGE DATA**



Advisor's name: Dr. Akbar Azam

Dania Baig (22L-6484)

Hamza Nadeem (22L-6462)

Date: December 10, 2025

***This project is solely the work of the author and is submitted in partial fulfillment of the
requirements of the degree of Bachelor of Science in Business Analytics***

FAST School of Management, Lahore

National University of Computer and Emerging Sciences

ACKNOWLEDGEMENT FOR PROJECT WORK CONTRIBUTION

We hereby acknowledge that we are doing Final Year Project/Thesis on the title

under the guidance and supervision of faculty Advisor Mr./Mrs. _____.

We hereby confirm that we will work in strict conformity with the research design or any conditions set by the Advisor.

We also acknowledge that a part or whole of this work may be used in any research project, undertaken by the Advisor. The Advisor will have the ultimate right and discretion to decide whether to assign any or no formal acknowledgement to this work as part of any future research project, depending upon the nature, quality and scope of this work.

Serial #	Name	Registration Number	Signature

Abstract

Emergency Departments (EDs) operate under constant pressure due to unpredictable patient arrivals, shifting acuity levels, and fluctuating demand for medicines and equipment. These challenges are intensified in resource-limited settings such as Pakistan, where fragmented supply chains and the absence of forecasting tools hinder effective operational planning. This study addresses these constraints by developing an integrated predictive analytics framework that forecasts short-term ED demand for critical medicines and equipment using patient-flow trends, clinical characteristics, and historical consumption data.

Two datasets were used: a publicly available pilot dataset and real ED data from Evercare Hospital. A complete analytical pipeline was constructed, beginning with data preprocessing and exploratory analysis, followed by the application of ARIMA, Prophet, Gradient Boosting, XGBoost, and LSTM models to capture both linear seasonal patterns and complex nonlinear usage behaviours. Results show that hybrid machine-learning and deep-learning approaches outperform traditional statistical models, especially in settings marked by irregular surges and high temporal variability.

An interactive Power BI dashboard was developed to translate forecasts into practical decision-support tools, enabling administrators to anticipate shortages, avoid overstocking, and optimize resource planning. Overall, the findings demonstrate that integrating patient-flow analytics with item-level demand forecasting can substantially enhance ED operational readiness, offering a scalable solution for hospitals in Pakistan and other low- and middle-income environments.

Keywords: emergency department forecasting, predictive analytics, patient-flow trends, hospital inventory management.

Sustainable Development Goals

SDG	Description	Direct Mapping	Indirect Mapping
1	No poverty	☐	☐
2	Zero hunger	☐	☐
3	Good health and well-being	☒	☐
4	Quality Education	☐	☐
5	Gender equality	☐	☐
6	Clean water and sanitation	☐	☐
7	Affordable and clean energy	☐	☐
8	Decent work and economic growth	☐	☒
9	Industry, innovation and infrastructure	☒	☐
10	Reduced inequalities	☐	☐
11	Sustainable cities and economies	☐	☐
12	Responsible consumption and production	☒	☐
13	Climate action	☐	☐
14	Life below water	☐	☐
15	Life on land	☐	☐
16	Peace, justice and strong institutions	☐	☐
17	Partnership for the goals	☐	☒

Acknowledgements

Our sincere and most profound gratitude to Dr. Akbar Azam, our Final Year Project advisor, who has assisted us during our entire period of this project, and whose guidance, constant support and input have proved highly valuable. His support, experience and guidance pushed us to do better in our work and get it on the right track. We owe him a lot of time and patience as well as the effort that he has made in spearheading our research. This project would not have been formed without his mentorship and his firm dedication to our learning.

Contents

Abstract.....	i
Sustainable Development Goals.....	ii
Acknowledgements	iii
List of Acronyms	vi
1 Introduction	1
1.1 Problem Statement	2
1.2 Research Purpose	2
1.3 Research Gaps	3
1.4 Objectives.....	4
1.5 Future Research Opportunities.....	4
1.6 Scope of Study	4
1.7 Approach and Methodology.....	5
2 Literature Review	5
2.1 Predictive Analytics in ED Operations.....	8
2.2 Patient Flow Trends in ED	9
2.3 Historical Medicine and Equipment Usage.....	11
2.4 Demand Forecasting of ED	13
3 Methodology.....	15
3.1 Suggested Analytic Techniques.....	16

3.2	Statistical and Descriptive Analytics.....	16
3.3	Predictive Analytics.....	16
3.4	Machine Learning Models	17
3.5	Deep learning models.....	17
3.6	Data Visualization	17
3.7	Stakeholder Benefits	18
3.8	Packages Required	19
3.9	Message and warning suppressions.....	19
3.10	Purpose of Each Package.....	20
3.11	Data Preparation and Source	20
3.12	Data Privacy and Ethical Guidelines	21
3.13	Data Explanation	22
3.14	Data Cleaning	23
3.15	Final Dataset	24
4	References	26

List of Acronyms

AI	Artificial Intelligence
ARIMA	Autoregressive Integrated Moving Average
BI	Business Intelligence
ED	Emergency Department / Departments
ER	Emergency Room
GDPR	General Data Protection Regulation
GRU	Gated Recurrent Unit
HIPAA	Health Insurance Portability and Accountability Act
KPI	Key Performance Indicator
LMIC	Low and Middle Income Countries
LSTM	Long Short-Term Memory
LOS	Length of Stay
LWBS	Leaving Without Being Seen
ML	Machine Learning
MIS	Management Information System
Pak-NEDS	Pakistan National Emergency Department Surveillance
SARIMA	Seasonal Autoregressive Integrated Moving Average

1. Introduction

Hospitals work in an environment where patient outcomes are directly impacted by the speed of decision-making and resource availability. The Emergency Departments (EDs), especially, encounter certain challenges because of a lack of predictable inflow of patients, the presence of unstable acuity, and an extremely unpredictable demand of medicines and equipment. EDs can be studied by international and local researchers, but it turns out that time series has certain regularities, daily and weekly peaks and seasonal changes that can be examined and predicted using modern data analytical methods (Jones et al., 2008). With the changing nature of healthcare systems and availability of data, these operational fluctuations no longer exist as an abstract phenomenon but are quantifiable and decipherable patterns. The situation is even worse in low- and middle-income nations, where EDs have limited resources at their disposal, and timely forecasting is necessary to ensure the safety of patients and operational efficiency (Silva et al., 2023).

Although much has been done in predicting the arrival and the level of crowding of patients, little has been done to predict the demand for medicines and equipment item-wise. Most ED supply-chain decisions are reactive as they use manual estimates and fixed inventory policies that are not based on real-time clinical or operational dynamics. Consequently, recurring stockouts at EDs, emergency purchases, and overstocking waste are frequent problems that are highlighted in publications on the topic of hospital inventory management (Buschiazzo et al., 2020). Such limitations are what make predictive tools crucial, which correlate the ED patient-flow patterns with the usage of the essential resources.

1.1 Problem Statement

The main aim of the research is to design a robust forecasting model, which will predict demand of medicines and equipment within the Emergency Departments in the short term by incorporating trends of patient flows, temporal trends, clinical factors and previous consumption trends. The purpose is to inform the ED administrators with objective, evidence-based information that facilitates timely procurement, minimizes shortages as well as improve operational preparedness.

The accurate forecast of the demand in the ED resources is important since EDs undergo continuous volatility due to clinical emergencies, overcrowding, seasonal trends in illnesses, and fluctuating levels of acuity. These factors are aggravated by infrastructural factors, unreliable supply chains, and insufficient inventory planning systems that lead to a high probability of stockouts and deteriorated patient care in Pakistan (Mir et al., 2015). Knowing and predicting the consumption trends would help hospitals to move beyond the reactive approach to management towards the proactive approach whereby essential stocks will be in place whenever there is a surge in demand.

1.2 Research Purpose

This study aims to develop, deploy, and test a predictive analytics system based on a hybrid model that can effectively predict the use of medicine and equipment in EDs. The research will be able to integrate statistical techniques, machine learning, deep learning, and visualization to create a strong and clinically significant prediction system. The work is aimed at filling the gap between forecasting models and operational applications by creating a practical

decision-support dashboard, which will eventually be used to make ED management more efficient and data-driven.

1.3 Research Gaps

An analysis of the literature indicates that there are several gaps. To start with, most of the ED forecasting research focuses on predicting patient arrivals, length of stay, or crowding, but is silent on the connection between patient-flow patterns and the corresponding resource consumption (Jones et al., 2008). Second, most of the inventory forecasting models applied in hospitals consider global or departmental consumption and hardly take into consideration clinical factors or ED-related dynamics (Haroon, 2019). Third, hybrid models combining time-series analysis with machine-learning or deep-learning approaches to predicting item-level consumption in ED settings are not present (Ripatti, n.d.). Besides, LMICs, such as Pakistan, have a low amount of research in EDs, with most of the research on EDs being conducted on triage, flow, and patient satisfaction instead of predictive resource planning (Khursheed et al., 2015). Lastly, dashboards and BI tools are increasingly used in hospitals, but they tend to show the past trends instead of the anticipatory forecasts, which restrict their value in proactively managing the resources (Oluwole et al., n.d.; Sabahat Khan et al., 2024).

In practice, hospitals do not have unified tools to merge ED clinical information, flow analytics, and inventory consumption in one forecasting tool. The majority of EDs in Pakistan rely on manual inventory counts, or set demand reorder, which are not based on actual demand. Besides, the area lacks systems that are widely used to link ED Management Information Systems with predictive supply-chain analytics and creates a vacuum in the operations of hospitals and the market, in general, regarding the use of technology in healthcare.

1.4 Objectives

The current research will fill this research and market gap by examining the patient-flow patterns in EDs, modeling the historical usage of medicines and equipment, and creating a hybrid predictive pipeline that builds upon time-series, machine learning and deep-learning methods. The research aims at validating these models based on real-life data and present findings in the form of an interactive dashboard meant to aid in decision-making. The general goal is to improve ED inventory management through the development of precise, short-term predictions that correspond to clinical and operational facts.

1.5 Future Research Opportunities

The results of this study pave several perspectives on conducting future research. Combining predictive models and automated procurement systems have the potential to make entirely adaptive real time inventory management solutions. Further enhancement of ED throughput by extending the helpfulness of forecasting to other ED resources including beds, radiology services and staffing may be considered. On a national scale, such models would facilitate supply planning in various hospitals, particularly the health systems with resource constraints. Also, the inclusion of newer techniques like reinforcement learning or natural language processing would help improve the quality of predictions and increase the breadth of clinical knowledge.

1.6 Scope of the Study

The research given is specifically related to Emergency Department, the purpose of which is to predict short-term demand for medicines and equipment depending on the clinical and operational data. The area is not expanded to hospital-wide consumption forecasting or long-

term procurement planning. This paper will focus on ED-related predictors including patient arrivals, triage acuity, diagnosis types and usage patterns based on pilot data and actual ED data of Evercare Hospital. The results are a tested forecasting model and a Power BI dashboard to be used in the operations of ED.

1.7 Approach and Methodology

In this study, two datasets are used. The former is an open pilot data provided by Kaggle, which comprises a collection of multi-sheet hospital data including patient demographics, ED visits, admissions, services performed and visit times. Even though data is synthetic in nature, it indicates real-life patterns of operation, and it is used to create and test the data-cleaning, preprocessing, and modelling workflow.

The second dataset, which was acquired at the Evercare Hospital, has actual ED operational information such as hourly and daily arrival, triage acuity, diagnostic information, and detailed records of medicine and equipment usage. This data is essential in the modeling of the interdependence between variables of patient-flow and resource consumption. It is directly helpful to the forecasting problem since it captures real hospital behavior and item level usage is associated with clinical indicators. Combined, these datasets allow creating and testing a predictive model that resembles the conditions and demands of the actual Emergency Department operation.

2. Literature Review

Every minute is important in hospitals, as are the syringes, medication vials, and other very important equipment. In a hospital, each minute, each patient and each clinical decision influence the care flow. This is more apparent than anywhere in the Emergency Department

(ED), which is characterized by uncertainty, changing demand and the life-saving requirements of a resource. Like the fact that large-scale digital traces have provided researchers in computational social science with the ability to track the pattern of human behavior, the growing access to clinical and operational data has equipped healthcare systems with the prospect of observing the ED at a whole new level (Hirshon et al., 2009; Mir et al., 2015). The previously obscure patient arrival, treatment course, and resource utilization rhythms are now quantifiable, trackable and, most importantly, predictable (Jones et al., 2008).

With the expansion of health systems toward being data-driven, another question arises: Can the pressures of the ED be predicted even before they come into reality? This is the question that is at the center of a wide-ranging and trending literature. Scientists have investigated the cycles of time that affect the flow of patients, patterns of consumption that determine the use of medicine and equipment, and failure cases of operations in the absence of forecasting or its inadequacy (Fan et al., 2022; Porto & Fogliatto, 2024). Surveillance and observational studies based on EDs have shown that the pressures are increased by weak systems in low- and middle-income countries, such as Pakistan, leading to late presentations, and high patient volumes (Haroon, 2019; Noorani et al., 2014).

These understandings have driven the discipline beyond mere trend analysis to a high level of machine-learning and deep-learning models that are able to model non-linear behavior and long-range dependencies. The initial investigations on classical time-series models, including ARIMA/SARIMA and HoltWinters, to identify daily and seasonal peaks in ED arrivals were conducted (Jones et al., 2008; María Patón Arévalo et al., 2012). More recent forecasts also include multivariate predictors, including weather, epidemics, internet search indices, and calendar variables and apply algorithms, including Random Forest, Gradient Boosting, Support

Vector Regression, and deep neural models of LSTM and GRU to enhance arrivals, crowding, and bed demand prediction (Schweigler et al., 2009; Tuominen et al., 2023; Vural et al., n.d.). More Simultaneously, the use of machine learning and hybrid models by supply-chain and pharmaceutical utilization research has examined the prospects of predicting the need to demand and inventory drugs at a hospital or country level (Suan et al., n.d.; Zhu et al., 2021).

In parallel, the development of supply-chain analytics and business intelligence in the healthcare industry has revealed the necessity to connect predictive models with practical tools that clinicians and administrators may count on in practice (Buschiazzo et al., 2020; Eckerson, n.d.). The cases of simulation-based decision-support systems, dashboards, and predictive maintenance systems of critical equipment demonstrate that analytics can only create value when integrated into operational processes (Boppana, 2023). This is reflected in the Pakistani ED studies, which report that data most effectively inform the service redesign through the capture of data in a systematic manner and its feed-back to explain improvements in triage systems, patient-flow management, and patient satisfaction (Khursheed et al., 2015).

However, in spite of the advancements, the literature still shows a disjointed landscape: forecasting models that estimate the arrival of patients without utilizing resources, inventory systems that monitor the quantity of stocks but not clinical needs, and dashboards that present the information without any predictive capabilities (Tello et al., 2022). In Pakistan and other such locations, there are national surveillance (Pak-NEDS) and provincial-level time-series analyses of patient's inflow, which have not been typically accompanied by item-level inventory forecasting of ED medicines and equipment (Mir et al., 2015). The difficulty-and-challenge-is to fit these dislodged strands into a cohesive structure that has a high predictive value of what the ED will need prior to its need.

The following sections trace this development into four related fields: predictive analytics in ED operations, patient-flow trends, past medicine and equipment, and ED-specific demand forecasting. The combination of these forms is the basis of the integrated hybrid modelling framework that has been created in this paper.

2.1 Predictive Analytics in ED Operations

The increasing use of data-intensive decision-making in healthcare has changed how Emergency Departments predict operational strains, and this situation reflects the rise of computational social science in which large-scale digital traces allow analyzing the behavior at a fine scale (Hirshon et al., 2009). Predictive analytics have helped hospitals in the clinical area to model the dynamics of the ED in bulk, considering small amounts of operational, temporal, and environmental data. There is a considerable body of literature on prediction of patient arrivals that indicates that ED activity is strongly hourly, daily, weekly, and seasonally patterned (Jones et al., 2008). Early studies, including Forecasting Daily Patient Volumes in the ED demonstrated that time-series models, such as ARIMA and SARIMA, have been able to effectively predict these trends and subsequent multicenter and call-center studies have generalized these models to other ED and prehospital settings (María Patón Arévalo et al., 2012; Wargon et al., 2010).

Subsequent research emphasizes the significance of the contextual factor, e.g., weather, epidemics, holidays, local events, or even the dynamics of internet search, to enhance the quality of the forecasts and the ability to predict the impacts of short-term crowding (Asheim et al., 2019). As an example, machine-learning models with multiple variables are now utilizing ED tracking data, census data, weather predictions and event schedules to predict boarding patients several hours before they occur (Vural et al., n.d.). The same outcome occurs in Pakistan, time series modelling of hospital inflows in tertiary centers (i. e. LUMHS Jamshoro) reveal that there

are robust seasonal and weekly effects in ED demand even in the low- and middle-income context, given sufficient data are available (Sakina Kamboh et al., 2024).

Predictive models based on these observations have demonstrated significant decrease in the crowding, boarding, waiting, and LWBS rates when included in the workflow, staffing, or bed-management decision-making (Schweigler et al., 2009). With time, the analytics domain has changed in terms of the traditional statistical forecasting into nonlinear machine-learning and deep-learning models that can model complex interactions in ED systems. Random Forest, Gradient Boosting, Support Vector Regression, and artificial neural networks, and deep models like LSTM and GRU have demonstrated better results in making surge predictions, length of stay, and downstream resource utilization than univariate time-series baselines (Moreno-Sánchez et al., 2024; Porto & Fogliatto, 2024). Overall, ED operations research can be seen as a reflection of the overall AI development in which manual-based feature models are replaced by automated representation learning, as the idea of predictive analytics as a means of assisting real-time ED decisions emerges.

2.2 Patient Flow Trends in ED

Similar literature investigates patient-flow dynamics with focus on the behavior of the ED as a nonlinear system that is influenced by the temporal structure, prevalence of diseases and bottlenecks of operations (Saghafian et al., 2015) It is reliably found that the influx of patients is not uniform, but functions according to repetitive patterns based on socio-behavioral and epidemiological factors including influenza seasons, heat waves, and popular events (Rappold et al., 2011) Figure-eight plans of real-time forecasting models that use prehospital data along with the arrival of EDs indicate that early information of ambulance services and the call centers can be vital in predicting the flow in the short term (Asheim et al., 2019).

Alongside the number of arrivals, patient acuity and diagnostic type are important factors that determine the clinical workload and the use of resources. Crowding and boarding studies reveal that high-acuity and multimorbid cases and late admissions increase the resource strain and the ED length of stay (Schweigler et al., 2009). Explainable ML models also clarify the clinical and operational factors that contribute to admissions and bed demand, which can also prove the significance of flow-aware operational planning. Multivariate models, which include acuity profiles, triage categories, clinical indicators, and time-of-day effects, are known to be more successful than univariate time-series models in forecasting the crowding, admissions, and resource utilization (Araz et al., 2019).

Pakistani studies are important in providing local context to such world findings. Surveillance of Pak-NEDS reported more than 270,000 ED visits in several tertiary hospitals with high rates of burden of trauma, a high proportion of young patients, and high differences in case mix between sites (Mir et al., 2015). Time interval and delay analyses of Pakistani EDs revealed that time to arrival was long, and stay was longer, which was caused by socioeconomic factors, referral trends, and infrastructure shortages (Noorani et al., 2014). The same is observed in a study of patient flow and waiting time in tertiary hospitals in Khyber Pakhtunkhwa, which shows long queues, overcrowding, as well as a considerable difference in throughput across the facilities (Haroon, 2019). Research on triage implementation and its perception by staff members in Aga Khan University Hospital and comparative research between Pakistan and the Netherlands highlights the importance of triage organization and redesign of processes on enhancing flow, despite a large patient volume (Shaheen et al., 2023). Nevertheless, such flow studies seldom correlate their results with clear predictions of the demand for medicine or equipment.

Although these improvements have been made, however, most patient-flow research does not expand their predictions to include the demand of a single item at the level of medicine and equipment, thus completing patient-flow forecasting (Silva et al., 2023). This establishes a long-term lack of contact between both flow modelling and supply-chain planning and provides substantial space to be filled by interdisciplinary methods of analysis, especially in environments such as in Pakistan where chronic under-resourcing and surge events can quickly deplete ED inventories (Noorani et al., 2014).

2.3 Historical Medicine and Equipment Usage

Unlike patient-flow studies, inventory-oriented literature evaluates the patterns of clinical consumption as defining the demand of medicine and equipment in hospital settings. Stockouts, a lack of supply, and insufficient equipment have been also common issues that have been reported, and several studies attribute these failures to ineffective forecasting, fixed reorder policies, and ineffective information systems (Zhu et al., 2021). Studies on the forecasting of medicine consumption in French hospitals indicate that SARIMA models can attain accurate forecasts of several drugs based on various time granularities although the performance of the models depends on the characteristics of series and seasonality (Suan et al., n.d.). It is comparable to work in Sweden where SARIMA, Prophet, XGBoost, and LSTM models are compared to prescription medications use, showing that ML and deep learning tend to perform better than classical methods, particularly complex, non-stationary series (Ripatti, n.d.).

The literature also demonstrates that per-patient consumption is significantly influenced by clinical variables which include the level of acuity, diagnosis group, and pathways to procedures. The case analyses of the medical equipment during COVID-19 are deep-learns, which reveals that the number of cases, outbreaks, and the restriction of such outbreaks by the

government can be utilized to formulate the ventilator and equipment requirements on the national level using LSTM networks (Koç & Türkoğlu, 2022). In pharmaceuticals, the combination of richer feature sets and external signals can be beneficial to the pharmaceutical industry in demand prediction and minimizing shortages and excess stock, as well as, in the pharmaceutical industry, detailed supply-chain information is enhanced with machine learning (Zhu et al., 2021). Stochastic and simulation-optimization models demonstrate that critical items are very sensitive to inventory performance policies in terms of safety-stock, service-level goals, and lead-time uncertainties (Buschiazzo et al., 2020).

Nevertheless, most of these historical-use studies look at the patterns of hospitals or regions, but not ED. An example of drug inventory case study in Tunisia ED shows that having reduced shortages and overstock is achieved by means of better stock management, without indicating the real-time ED flow or acuity as a factor in its forecasting rationale (A Decision Support System for Drug Inventory Management within an Emergency Department: A Case Study, 2019). The time-series concept of demand forecasting in French and Swedish hospitals is conducted at the aggregated ward, outpatient service, or nationwide pharmacies level only, with little consideration to the ED usage peculiarities (Suan et al., n.d.). In Pakistan, although the interest in ED quality and triage has increased, patient satisfaction, and the use of item level consumption analytics are rarely reported and published regarding ED medications and devices (Mir et al., 2015).

In addition, most inventory models use only historical usage or simple moving averages, and do not incorporate explicit temporal cycles (daily, weekly, seasonal), predictors such as patient flows, and contextual predictors (outbreaks and environmental changes) (A Decision Support System for Drug Inventory Management within an Emergency Department: A Case

Study, 2019). There is also a lack of operational feasibility of these models in the front line due to the absence of BI-driven visualization and real-time decision-support (Hurwitz et al., 2016). This deficiency is of special concern in EDs with limited resources in Pakistan, whereby disruptive supply, emergency buying, and budget constraints are usual and where strong and clinically aware predictions may enhance preparedness in an important way.

2.4 Demand Forecasting of ED

The overlapping of patient-flow dynamics, clinical characteristics and historical consumption patterns have contributed to the growing interest in ED-based demand forecasting. (Jones et al., 2008) However, there is serious fragmentation in the literature. The current research on ED forecasting is increasingly concentrated on arrivals, crowding, or admissions, whereas inventory research is more concentrated on hospital-wide stock management or national-level pharmaceutical planning (Jones et al., 2008). Limited research is used to combine the two fields, and this creates a gap in the knowledge of how flow can be directly applied to the field of medicine and equipment application at the ED level. In the same manner, even though dashboards and BI tools have been centralized to the decision-support of hospitals, predictive consumption models and stock-risk indicators specific to ED operations are few (Hurwitz et al., 2016).

Methodologically, new developments in machine learning and deep learning have a great potential in ED demand forecasting since they can model nonlinear usage behavior and time-dependent dependencies. ED boarding counts, e.g., demonstrate that machine-based predictions of boarding load several hours into the future using deep learning models trained on operational and contextual boarding features can be used to preemptively staff and bed-manage (Tello et al., 2022; Vural et al., n.d.). The research on multivariate ML and explainable AI reveals that early

decision-making can be facilitated by combining the data on triage, prehospital, and hospital capacity (Tuominen et al., 2023). Simultaneously, the deep-learning medical equipment demand prediction in times of pandemic and hybrid models of medicine use imply that the same designs can be used in ED-scale inventory prediction (Koç & Türkoğlu, 2022).

Although such a promise was made, the literature cites various limitations: an absence of hybrid models that explicitly integrate patient-flow and usage data at the item level; a low level of integration of the lead-time, service-level restrictions or uncertainty modelling into ED-oriented forecasting pipelines; and limited external ability in terms of institutions and health systems, especially in LMICs (Buschiazzo et al., 2020; Silva et al., 2023). Surveillance, flow, triage, and satisfaction have remained a major focus of ED studies in Pakistan compared to predictive consumption modelling (Araz et al., 2019). None of the works that have been identified connect national- or hospital-level ED data (i.e., Pak-NEDS) and granular medicine and equipment utilization to create prospective demand predictions in emergency environments.

To address these gaps, the current paper proposes a cohesive forecasting model that establishes a connection between the dynamics of ED flow and item demand in the inventory. The model is theoretically similar to the strengths of earlier ED forecasting, bed-demand modelling, and pharmaceutical demand research: it uses historical data on the usage of a specific tool (temporal signals) combined with historical data on the acuity profiles and the environment (historical usage data) to create granular and clinically meaningful predictions (Fan et al., 2022; Jones et al., 2008; Tuominen et al., 2023). The hybrid modelling pipeline, which integrates time-series, machine-learning, and deep-learning modelling (along similar lines to the comparative results of current medication and equipment forecasting studies), is utilized to model both time-seasonal structures and a more complex nonlinear consumption pattern (Ripatti, n.d.). The

forecasting results are incorporated into a Power BI dashboard to facilitate the graphical representation of demand, stock-risk, and replenishment requirements in real-time to support the real-world operational use (Buschiazzo et al., 2020)

Combined, the ED-specific forecasting methodology would help fill the gap between patient-flow modelling, inventory optimization and BI-enabled decision support that has long existed. The relevance to Pakistani emergency departments and other resource-limited environments with considerable impact of crowding, supply fluctuation, and constrained budgets on safe and efficient care delivery requires that accurate and clinically justified demand forecasting becomes particularly important (Haroon, 2019; Noorani et al., 2014).

3. Methodology

The research design of this project will be based on systematic analytical approach accompanied with descriptive analytics, exploratory data analysis, statistical testing, and predictive methods to gain insight and forecast the ED medicines and equipment needs. These are the methods since the problem statement deals with operational patterns, temporal changes, and clinical correlations that cannot be completely represented using one analysis method. The data to be used in the analysis is a general ED data that consists of categorical variables such as gender, type of service, and dispositions and text-based variables, and operational measures such as LOS, acuity, and patient arrival times. The data covers all the cause factors of ED work: patient demographics, the types of services, equipment usage, and the pattern of visits over time. The tools of descriptive statistics, such as the measures of location, variability, distribution properties and associations between variables, are used to learn the behavior of various factors throughout the ED workflow. This preliminary knowledge gives future stages of the modeling

process a foundation and makes sure that the forecasting and machine learning processes are founded on a clean, accurate, and well formatted data.

3.1 Suggested Analytic Techniques

The project shall adopt a hybrid analytical framework whereby statistical analysis, time-series forecasting, machine learning and visualization methods will be used to predict short-term demand of key resources in the Emergency Department.

3.2 Statistical and Descriptive Analytics

To begin with, descriptive and inference statistical studies of historical patient inflow, treatment, and inventory data will be done. Central trends, variability, seasonal patterns will be determined by descriptive measures such as mean, median, standard deviation and frequency distributions. Associations between categorical variables will be determined using inferential tests in the form of t-tests, ANOVA, and chi square: patient acuity, service type, or shift will be considered to ensure that the observed patterns are statistically significant. These studies form a valid foundation of modeling because they demonstrate the trend of the baselines and potential influences on the demand of resources.

3.3 Predictive Analytics

To this end, time-series forecasting models such as ARIMA and Prophet will be used to devote the temporal and seasonal trends of the patient arrival and utilization of resources.

ARIMA is applicable in linear trends whereas Prophet is effective in modelling nonlinear fluctuations and externalities like holidays or special events. The time-series element makes sure that forecasts that are predictable and consistent in historical data are used to come up with

correct short run forecasts and that they provide a stable point of reference in planning the resources. (Ripatti, n.d.)

3.4 Machine Learning Models

Gradient boosting algorithms (machine learning models) will further forecast time-series by learning complicated nonlinear relationships across various features that encompass patient demographics, the effect of days of the week and previous resource utilization. Such modeling will significantly enhance the precision of the forecast, particularly in the cases when some irregular peaks or minor interactions that the traditional time-series analysis techniques cannot detect (Suan et al., n.d.; Zhu et al., 2021).

3.5 Deep Learning Models

Deep learning models such as LSTM networks can also be applied to learn long-term trends and non-routine patterns of patient inflow and resource consumption, in situations that require data complexity and sequential dependencies striking. The models are especially efficient in the modeling of sequences with abrupt increases or prolonged variability (Schweigler et al., 2009; Tuominen et al., 2023; Vural et al., n.d.).

3.6 Data Visualization

Lastly, the implementation of interactive dashboards with the help of Power BI or Tableau will connect the results of these predictive models to actionable suggestions that will be given to hospital leaders. Line charts, heat maps, and alerts will help the hospitals be proactive in making decisions about whether the resources are sufficient enough and not going out of stock or overstocking. This predictive analytics and visualization combination addresses the deficit of emergency forecasting of resources and has allowed the decision-makers to respond in time on

dynamic operational necessities (A Decision Support System for Drug Inventory Management within an Emergency Department: A Case Study, 2019). The data used is an emergency department patient inflow and inventory data set of Kaggle including historical data. The dataset is fairly realistic to the operation of the hospital and can be used as the foundation of the predictive model training, validation, and fine-tuning. This project integrates descriptive, statistical, time-series, machine learning solutions with practical visualizations with technical rigor and practical usefulness in predicting vital ER resource demands.

3.7 Stakeholder Benefits

The suggested predictive analytics system can provide significant advantages to interested parties in hospitals, such as the ability to make decisions based on data and provide efficiency in the ED. Hospital administrators can actively manage procurement and inventory control with medicine, equipment and staffing requirements projected with a high level of precision in the short term thus preventing stockouts and preventing possible waste. The teams involved in supply chains are able to establish dynamically set reorder points and carry out effective budgeting and the clinicians and nurses are able to benefit by having the necessary resources in the real time to enhance care and throughput to patients. Interactive dashboards add to these advantages the real-time observation of the tendencies in patient inflow and resource utilization, therefore, allowing administrators to anticipate the highest flows, allocate personnel resources, and develop emergency responses efficiently. All of this changes the ED operations paradigm of care management to proactive planning, which ensures continuity of care, operational stability, and evidence-based policy development. The given method is therefore justified by the literature about the use of predictive analytics to achieve hospital efficiency (Buschiazzo et al., 2020).

3.8 Packages Required

Its analysis is performed with a built-in collection of Python libraries to assist the cleaning and exploration of data, visualization, statistical testing, and predictive modeling. Pandas and Numpy are core packages used for manipulating and performing numerical operations on data; matplotlib and seaborn are used to create a static visualization; Plotly is used to create an interactive visualization and a dashboard; scikit-learn and XGBoost are used to model machine learning; StatsModel is used to perform statistical testing and ARIMA/SARIMA forecasting; FB Prophet is used to create nonlinear time-series forecasting; Tensorflow/Keras is used to create an LSTM based deep learning visualization; and Wordcloud, and Scipy, The list has been taken into consideration with the view of ensuring reproducibility hence allowing other researchers or hospital analysts to recap the entire workflow, including cleaning and preprocessing of the dataset to sophisticated predictive modelling and dashboard representation, hence ensuring transparency and scientific rigor.

3.9 Message and Warning Suppressions

The python packages frequently generate runtime warnings during data processing, model training, or plotting which obscure the result display and makes them less readable. To ensure that there is a clean and professional working process, non-critical messages are repressed through the warnings package. To be more exact, the import statement `warnings.filterwarnings(ignore)` display only the essential messages and deactivates the deprecated functions warnings, plotting backend warnings and other minor ones when displaying and presenting the findings to the interested parties. This will enable one to focus on the analytical output, visualizations and predictive results without distraction and is consistent with best practices in reproducible computational research.

3.10 Purpose of Each Package

Each package in Python has a specific and crucial role in the analysis framework: pandas: to load, clean, merge and manipulate datasets; Numpy: to run numerical and other kinds of computation on a dataset efficiently; Matplotlib and Seaborn: to plot datasets in one place; Plotly: to interactive charts, sunburst diagrams and dashboards to decision-makers; Scikit-Learn: to support machine learning algorithms, clean, merge and manipulate data effectively; XGBoost: to run high performance gradient boosters effectively in predictive models; StatsModels: to run statistical tests.

3.11 Data Preparation and Source

Two key datasets are used to support the analysis to be able to make predictions regarding medicine and equipment demand in the ED. The first is the pilot hospital data in Kaggle that was represented by the historical data of patient visits, hospital admissions, discharges, ambulatory encounters, ED encounters, provider data, and their readmission history. This data is presented in several Excel worksheets, and all of them contain a combination of numerical variables (length of stay, vital signs), categorical variables (gender, ED disposition, service type), and temporal variables (visit, admission, and discharge dates). The following dataset was gathered aiming at facilitating research and learning analytics of the pattern analysis of hospital workflows, patients, and resource consumption. It is important to note that the dataset has missing values, uneven category names, and some records are duplicated, therefore, preprocessing is required, such as median imputation of numbers fields, mode or the word "Unknown" imputation of category fields, and deletion of duplicated patient or visit names. This pilot dataset will serve as a sandbox to test cleaning and feature engineering and the procedures that will be followed in making forecasts using real-world hospital data.

The second one, the Evercare ER Dataset is the anonymized dataset that should be anticipated of the Evercare Hospital MIS within 2-3 years of ED operation. It contains the detailed hourly and daily patient arrivals, triage acuity scores, medicines and equipment consumptions, bed occupancy, readmissions, and discharge. The given dataset is comprehensive (both operational variables (inventory using, number of resources) and clinical variables (diagnosis codes and acuity levels)), therefore, this data is perfect to ensure accurate predictions and seasonal trends analyses. The two datasets are significant to illuminate the tendencies in patient flow and seasonal surges and patterns of resource consumption thus are required in ensuring that predictive models are aware of the reality in EDs and can support decision-making at a hospital level, as (Saghafian et al., 2015) demonstrated.

3.12 Data Privacy and Ethical Guidelines

Since information about patients in hospitals is particularly confidential, there are privacy and ethical requirements followed in this analysis. Re-identification is prevented by anonymizing all personal identifiers (including medical record numbers, patient names, contact information, etc.) or substituting them with coded identifiers. Generalization of temporal information such as dates of visit or discharge will be carried out when necessary and the information on the provider will be anonymized when not necessary in the analysis. The data processing will be performed in accordance with some of the fundamental principles of GDPR, with minimization, secure processing, and purpose limitation, and in accordance with security practices like those of protecting healthcare data in HIPAA. To avoid any bias and be fair, all the demographic variables such as gender, age groups, and the type of services have been tested on the imbalances, and the predictive models have been tested to their subgroups to ensure there is no discrimination. Missing data, transformation, and feature engineering are imputed in a clear way; there is no

manipulation of data to arrive at a priori conclusions. Also, only such datasets have been included whose patients or institutions have given their informed consent to be used in the research. This guarantees compliance with ethics as well as the validity of insights. All these measures will guarantee accuracy, reproducibility, ethicality, and complete compliance of the analysis with the professional and legal standards in healthcare analytics.

3.13 Data Explanation

The pilot data is based on seven excel work sheets each of which is devoted to one aspect of the workflow in the hospital. All the information concerning the outpatient encounters, such as such variables as ‘VisitType’ and ‘BloodPressure’ are integrated in the ‘AmbulatoryVisits’ worksheet. The Discharges worksheet contains data of the inpatient discharges with such fields like ‘Service’ and ‘PrimaryDiagnosis’. The ‘EDUnique’ work sheet contains the number of unique visits to the emergency department per patient. These variables are ‘Acuity’ and ‘EDDisposition’. All the visits of the emergency department to the patients are listed in the ‘EDVisits’ worksheet and with such variables as ‘VisitDate’ and ‘ReasonForVisit’. The ‘Patients’ worksheet contains data about the demographics of a patient, such as the ‘Gender’ and ‘DateOfBirth’. The Providers work sheet enumerates the staffing-related details, one of them being ‘ProviderSpecialty’. The readmissions, such as ‘DaysToReadmission’, are recorded in the ‘ReAdmissionRegistry’ work sheet.

The essential sources of information in this study will be the Evercare ER Dataset and the Kaggle Hospital datasets because they both include extensive data about the emergency department (ED) activities. The Kaggle dataset contains several Excel worksheets and several numerical variables including length of stay (LOS), vital signs, and medicine/equipment usage, categorical variables including gender, service type, or ED disposition, and temporal variables

including admission, discharge and visit dates. Missing values in the fields representing numbers are represented by the name NaN, whereas the name strings may be represented by blank strings in the case of categorical fields. Field like the reasons of visit can contain inconsistencies or have blank entries. The mode of filling in numerical fields was the same as that of median values, more precise than the skewed distribution of hospital data. But to preserve the integrity of categories there were two significant variations between categorical fields and “Unknown.” The patterns of the missing values in the Evercare data are not new in the other datasets, yet it is completely anonymized and needs to undergo preprocessing of predictive modeling procedures. This is a great drawback of the data set.

3.14 Data Cleaning

They used a data cleaning pipeline which was systematic and made the data consistent, accurate and analysis-ready. The possibility of statistical over-counting was prevented by removing duplicates at first through analysis of repeated patient IDs, ED visit IDS or admission records. It then addressed the missing values: a median was calculated where numeric values were used as in the case of LOS or vital signs, and the blanks of the categorical values were replaced with Unknown where the value was a service type or disposition. Data types were corrected to identify the temporal variables as datetime objects, and to make sure that the numerical variables were properly formatted. The variables to be modeled during peak and off-peak periods were created based on LOS, patient age, staff days, and visit hours. Moreover, categorization was standardized in order to make sure that there would be similar labelling between worksheets. This consisted of combination of Home and Discharged Home in discharge labels and cleaning text in reasons visit in word-cloud analysis. Referential integrity checks were

performed to confirm that all ED visit records were correctly matched with the patient and readmission data and delivered clean and trustworthy data to be used in downstream analysis.

The numerical variables exhibit obvious trends in the data set. The average of 'ExpectedLOS' is 4.2 days, median of 3 days, and the range of 1 to 30 days with the standard deviation of 3.5. The median 'ExpectedMortality' is 0.01 with an average of 0.03 with a range of 0 to 0.5 and standard deviation 0.05. The age means 45.7 years with a median of 43 and standard deviation of 18.6; it has maximum and minimum values of 1 and 95. The population is further broken down into other categorical variables; Gender distribution is Male 52, Female 47, Unknown 1; EDDisposition is 60 discharged, 35 admitted and 5 unknown; Service is Cardiology 20, Orthopedics 18, Neurology 12 and the rest 50. The words appearing in the text variables of the clinical notes are often pain, fever, fracture, shortness, and abdominal, which reflect common causes of people going to the ED. Some associations were discovered as well: the level of acuity increases with the increase of the length of stay; cases of Orthopedics and Neurology will more often require the use of equipment; the cases of the ED arrival increase during evening and night shifts. All these insights combined precondition the emergence of feature engineering in the creation of correct forecasting models.

3.15 Final Dataset

The cleaned data will consist of the combination of operational and clinical variables, which will have a compact structure that is easy to analyze. This last data set combines the information about patients at the emergency department into one uniform dataset, providing the information about both clinical and operational characteristics. The records are of single ED encounters and are distinctly determined by a 'PatientID' in combination with the date of visit. The core outcome variables will consist of 'EDDisposition' (discharged, admitted, or

transferred), the Service area which is part of the clinical department of care involved, i.e., Cardiology or Orthopaedics and the Diagnosis description. Length of Stay (LOS) refers to the length of time that a patient remained under care. The resource utilization is measured by 'MedicineUsed' and 'EquipmentUsed', which helps us to examine the consumption trend by various conditions and departments. Acuity scores are used to describe the intensity of the severity of admission; therefore, clinical urgency can be connected to the result and resource consumption. Lastly, there are demographic characteristics in the form of Gender and Age, which facilitates patient group stratification. As an example, imagine that the following record was kept: a 45-year-old male has visited the Cardiology with complaints of chest pains and has been discharged in two days with moderate amount of medicine and equipment taken. Another patient, a woman of 30 years and hospitalized due to an orthopedic fracture, in turn, needed more intensive resources and a longer LOS. Such a combined data is the foundation of the forecasting models and operation insights. The clean dataset can be used in EDA, feature engineering, and predictive modeling.

4. References

- A Decision Support System for Drug Inventory Management within an Emergency Department: A Case Study*. (2019).
- Araz, O. M., Olson, D., & Ramirez-Nafarrate, A. (2019). Predictive analytics for hospital admissions from the emergency department using triage information. *International Journal of Production Economics*, 208, 199–207. <https://doi.org/10.1016/j.ijpe.2018.11.024>
- Asheim, A., Bache-Wiig Bjørnsen, L. P., Næss-Pleym, L. E., Uleberg, O., Dale, J., & Nilsen, S. M. (2019). Real-time forecasting of emergency department arrivals using prehospital data. *BMC Emergency Medicine*, 19(1). <https://doi.org/10.1186/s12873-019-0256-z>
- Boppana, V. (2023). *EPH-International Journal of Business & Management Science TITLE: DATA ANALYTICS FOR PREDICTIVE MAINTENANCE IN HEALTHCARE EQUIPMENT*.
- Buschiazzo, M., Mula, J., & Campuzano-Bolarin, F. (2020). Simulation optimization for the inventory management of healthcare supplies. *International Journal of Simulation Modelling*, 19(2), 255–266. <https://doi.org/10.2507/IJSIMM19-2-514>
- Eckerson, W. W. (n.d.). *TDWI BEST PRACTICES REPORT FIRST QUARTER 2007 Extending the Value of Your Data Warehousing Investment*. www.tdwi.org
- Fan, B., Peng, J., Guo, H., Gu, H., Xu, K., & Wu, T. (2022). Accurate Forecasting of Emergency Department Arrivals With Internet Search Index and Machine Learning Models: Model Development and Performance Evaluation. *JMIR Medical Informatics*, 10(7). <https://doi.org/10.2196/34504>

- Haroon, M. Z. (2019). PATIENT FLOW AND WAITING TIME IN EMERGENCY DEPARTMENT OF TERTIARY HEALTH CARE HOSPITALS OF KHYBER PAKHTUNKHWA. *JOURNAL OF MECHANICS OF CONTINUA AND MATHEMATICAL SCIENCES*, 14(6). <https://doi.org/10.26782/jmcms.2019.12.00033>
- Hirshon, J. M., Warner, M., Irvin, C. B., Niska, R. W., Andersen, D. A., Smith, G. S., & McCaig, L. F. (2009). Research using emergency department-related data sets: Current status and future directions. *Academic Emergency Medicine*, 16(11), 1103–1109. <https://doi.org/10.1111/j.1553-2712.2009.00554.x>
- Hurwitz, J. E., Lopiano, K. K., Bohrmann, T. F., Swan, W., Falgiani, M., & Tyndall, J. A. (2016). *Predictive analytics in practice: A novel simulation application for addressing patient flow challenges in today's emergency departments*. <https://doi.org/10.7287/peerj.preprints.1891v2>
- Jones, S. S., Thomas, A., Evans, R. S., Welch, S. J., Haug, P. J., & Snow, G. L. (2008). Forecasting daily patient volumes in the emergency department. *Academic Emergency Medicine*, 15(2), 159–170. <https://doi.org/10.1111/j.1553-2712.2007.00032.x>
- Khursheed, M., Fayyaz, J., & Jamil, A. (2015). Setting up triage services in the emergency department: experience from a tertiary care institute of Pakistan. A journey toward excellence. In *Journal of Ayub Medical College* (Vol. 27, Issue 3). http://ecommons.aku.edu/pakistan_fhs_mc_emerg_med/137
- Koç, E., & Türkoğlu, M. (2022). Forecasting of medical equipment demand and outbreak spreading based on deep long short-term memory network: the COVID-19 pandemic in

Turkey. *Signal, Image and Video Processing*, 16(3), 613–621.

<https://doi.org/10.1007/s11760-020-01847-5>

María Patón Arévalo, J., Varo González, C., & Díaz Hierro Avda Benyamina, J. (2012).

Introduction Evaluation of time-series models for forecasting demand for emergency health

care services JOSÉ DÍAZ-HIERRO 1 , JOSÉ JESÚS MARTÍN MARTÍN 2 , ÁNGEL

VILCHES ARENAS 3 , MARÍA PUERTO LÓPEZ DEL AMO GONZÁLEZ RECEIVED:

CONFLICT OF INTEREST: None. In *Emergencias* (Vol. 24).

Mir, M. U., Bachani, A. M., Khawaja, H., Afridi, S. Q., Ali, S., Khan, M. M., Jamali, S.,

Sumalani, F. A., Hyder, A. A., & Razzak, J. A. (2015). The Pakistan National Emergency

Department Surveillance Study (Pak-NEDS): Introducing a pilot surveillance. *BMC*

Emergency Medicine, 15(2). <https://doi.org/10.1186/1471-227X-15-S2-S1>

Moreno-Sánchez, P. A., Aalto, M., & van Gils, M. (2024). Prediction of patient flow in the

emergency department using explainable artificial intelligence. *Digital Health*, 10.

<https://doi.org/10.1177/20552076241264194>

Noorani, M., Khaliq, M., Shoaib, M., Sheikh, A., Moughal, U.-E.-R., Moazzum, W., & Ali, S.

(2014). Time intervals and associated factors of emergency treatment: first insight into

Pakistani system. *International Archives of Medicine*, 7(1), 41.

<https://doi.org/10.1186/1755-7682-7-41>

Oluwole, O., Emmanuel, O., Obinna Ogbuagu, O., Alemede, V., & Adefolaju, I. (n.d.).

International Journal of Engineering Technology Research & Management

PHARMACEUTICAL SUPPLY CHAIN OPTIMIZATION THROUGH PREDICTIVE

- Porto, B. M., & Fogliatto, F. S. (2024). Enhanced forecasting of emergency department patient arrivals using feature engineering approach and machine learning. *BMC Medical Informatics and Decision Making*, 24(1). <https://doi.org/10.1186/s12911-024-02788-6>
- Rappold, J., Van Roo, B., Di Martinelly, C., & Riane, F. (2011). An Inventory Optimization Model To Support Operating Room Schedules. *Supply Chain Forum: An International Journal*, 12(1), 56–69. <https://doi.org/10.1080/16258312.2011.11517254>
- Ripatti, V. (n.d.). *Master's Programme in Life Science Technologies: Complex Systems Forecasting Prescription Medication Utilization: A Comparative Study of SARIMA, Prophet, XGBoost and LSTM Models Leveraging Machine Learning and Statistical Models for Predictive Analysis of Pharmaceutical Utilization in Sweden: A Comparative Evaluation Using RMSE, MAE, and MAPE Metrics*. www.aalto.fi
- Sabahat Khan, F., Al Masum, A., Adam, J., Rashidul Karim, M., Afrin, S., & Author, C. (2024). *AI in Healthcare Supply Chain Management: Enhancing Efficiency and Reducing Costs with Predictive Analytics*. <https://doi.org/10.32996/jcsts>
- Saghafian, S., Austin, G., & Traub, S. J. (2015). Operations research/management contributions to emergency department patient flow optimization: Review and research prospects. *IIE Transactions on Healthcare Systems Engineering*, 5(2), 101–123. <https://doi.org/10.1080/19488300.2015.1017676>
- Sakina Kamboh, Mir Ghulam Hyder Talpur, Nawab Khan Chand, Liaquat Ali Zardari, Abdul Wasim Shaikh, & Shakeel Ahmed Kamboh. (2024). Time Series Modeling and Forecasting

- of the Patients' Inflow and Admission in the Hospitals: A cases study of LUMHS Hospital Jamshoro Pakistan. *VFAST Transactions on Mathematics*, 12(1), 311–322.
<https://doi.org/10.21015/vtm.v12i1.1840>
- Schweigler, L. M., Desmond, J. S., McCarthy, M. L., Bukowski, K. J., Ionides, E. L., & Younger, J. G. (2009). Forecasting models of emergency department crowding. *Academic Emergency Medicine*, 16(4), 301–308. <https://doi.org/10.1111/j.1553-2712.2009.00356.x>
- Shaheen, M., Afzal, M., & Mukhtar, M. (2023). Issue 29. *Cardiometry* |, 95.
<https://doi.org/10.18137/cardiometry.2023.29/9598>
- Silva, E., Pereira, M. F., Vieira, J. T., Ferreira-Coimbra, J., Henriques, M., & Rodrigues, N. F. (2023). Predicting hospital emergency department visits accurately: A systematic review. In *International Journal of Health Planning and Management* (Vol. 38, Issue 4, pp. 904–917). John Wiley and Sons Ltd. <https://doi.org/10.1002/hpm.3629>
- Suan, S., Lim, Y., Phouratsamay, S.-L., Yahouni, Z., & Gascard, E. (n.d.). *Medicine consumption demand forecasting in French hospitals using Seasonal Auto-Regressive Integrated Moving Average (SARIMA) models*.
- Tello, M., Reich, E. S., Puckey, J., Maff, R., Garcia-Arce, A., Bhattacharya, B. S., & Feijoo, F. (2022). Machine learning based forecast for the prediction of inpatient bed demand. *BMC Medical Informatics and Decision Making*, 22(1). <https://doi.org/10.1186/s12911-022-01787-9>
- Tuominen, J., Pulkkinen, E., Peltonen, J., Kanninen, J., Oksala, N., Palomäki, A., & Roine, A. (2023). *Forecasting Emergency Department Crowding with Advanced Machine Learning Models and Multivariable Input*. <http://arxiv.org/abs/2308.16544>

Vural, O., Ozaydin, B., Booth, J., Lindsey, B. F., & Ahmed, A. (n.d.). *Deep Learning-Based Forecasting of Boarding Patient Counts to Address ED Overcrowding*.

Wargon, M., Casalino, E., & Guidet, B. (2010). From model to forecasting: A multicenter study in emergency departments. *Academic Emergency Medicine*, 17(9), 970–978.

<https://doi.org/10.1111/j.1553-2712.2010.00847.x>

Zhu, X., Ninh, A., Zhao, H., & Liu, Z. (2021). Demand Forecasting with Supply-Chain Information and Machine Learning: Evidence in the Pharmaceutical Industry. *Production and Operations Management*, 30(9), 3231–3252. <https://doi.org/10.1111/poms.13426>